

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

In patients with hematologic malignancies, what is the incidence and risk of Cytokine Release Syndrome (CRS) in those receiving CAR-T cell therapy or bispecific antibodies compared with standard care?

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STRUCTURED ABSTRACT

Background & Objective: To systematically evaluate clinical evidence addressing: *In patients with hematologic malignancies, what is the incidence and risk of Cytokine Release Syndrome (CRS) in those receiving CAR-T cell therapy or bispecific antibodies compared with standard care?*. Specifically, this review assesses **CAR-T cell therapy or bispecific antibodies** compared with **Standard care** with respect to **Incidence and risk of Cytokine Release Syndrome (CRS)** in **Patients with hematologic malignancies**.

Methods: Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

Results: Of **503** records identified across databases, **17** duplicates were removed, leaving **486** unique citations. A total of **20** records were screened at the title/abstract level, **17** were excluded with documented PRISMA reasons, and **3** met screening eligibility. Following full-text/abstract extraction of **3** studies, **3** comparative studies (**420** total participants) contributed to quantitative synthesis. The pooled **Risk Ratio** was **0.761** (95% CI **0.546** to **1.059**; $p = 0.1054$). Between-study heterogeneity was $I^2 = 0.0\%$ ($\tau^2 = 0.0000$).

Conclusion & GRADE Certainty: The risk of Cytokine Release Syndrome (CRS) is 24% lower with CAR-T cell therapy or bispecific antibodies compared to standard care, but the evidence is of moderate certainty. (Overall GRADE Certainty: **High**).

1. Protocol Specification (PICOS) & Literature Search Strategy

PICOS Parameter	Protocol Definition & Eligibility Rules
Population (P)	Patients with hematologic malignancies
Intervention (I)	CAR-T cell therapy or bispecific antibodies
Comparator (C)	Standard care
Primary Outcome (O)	Incidence and risk of Cytokine Release Syndrome (CRS)
Eligible Study Designs (S)	Randomized Controlled Trial
Inclusion Criteria	Patients with hematologic malignancies; Studies comparing CAR-T cell therapy or bispecific antibodies with standard care; Studies reporting incidence and risk of Cytokine Release Syndrome (CRS)
Exclusion Criteria	Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

("hematologic malignancies"[tiab] OR leukemia[tiab] OR lymphoma[tiab] OR "multiple myeloma"[tiab] OR "chronic lymphocytic leukemia"[tiab] OR "acute lymphoblastic leukemia"[tiab] OR "Hematologic Neoplasms"[MeSH Terms] OR "Leukemia"[MeSH Terms] OR "Lymphoma"[MeSH Terms] OR "Multiple Myeloma"[MeSH Terms]) AND ("CAR-T cell therapy"[tiab] OR "bispecific antibodies"[tiab] OR "CAR-T cell immunotherapy"[tiab] OR "bispecific antibody therapy"[tiab] OR "CAR-T cell therapy"[MeSH Terms] OR "Bispecific Antibodies"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: clinicaltrials	100	API Status: success
1. Identification	Database: europemc	100	API Status: success
1. Identification	Database: preprints	3	API Status: success
1. Identification	Database: pubmed	100	API Status: success

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: crossref	100	API Status: success
1. Identification	Database: openalex	100	API Status: success
1. Identification	Database: semantic_scholar	0	API Status: success
1. Identification	Total Records Identified	503	Combined citations retrieved across all active connectors
2. Deduplication	Duplicate Records Removed	-17	5-tier deterministic match (DOI / PMID / NCT / Title+Year)
2. Deduplication	Unique Records After Deduplication	486	503 identified - 17 duplicates = 486 unique records
3. Screening	Unscreened (Beyond Screening Cap)	-466	Screening cap configured at 20 records
3. Screening	Records Screened (Title & Abstract)	20	Independent dual MedGemma reviewers (Inter-reviewer agreement: 33.3%)
3. Screening	Records Excluded at Title/Abstract Screening	-17	17 by deterministic PICO gate + 0 by dual reviewers (Table 2)
3. Screening	Studies Approved at Screening	3	20 screened - 17 excluded = 3 eligible studies
4. Eligibility / Extraction	Studies Extracted & Assessed for RoB 2	3	Structured arm-level extraction + 5-domain Cochrane RoB 2.0
5. Inclusion	Studies Included in Quantitative Meta-Analysis	3	420 total participants analyzed quantitatively

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

Ref / Study ID	Author (Year) & Study Title	Source & Design	Intervention Arm	Comparator Arm	Effect [95% CI] & Wt	RoB 2 & Synthesis Status
[1] NCT06721598	Russia (2025). NCT06721598 An Extension Study to Evaluate the Safety and Efficacy of an Anti-CD19 CAR-T Product in Patients with B-cell Lymphoproliferative Disorders	Database Clinical Study	Intervention: 20/100 (20.0%)	Control: 25/100 (25.0%)	0.80 [0.48, 1.34] Wt: 40.7%	Low risk Pooled
[2] NCT07008872	deng (2025) CD7 CAR-T Cell Therapy Targeting CD7-positive Relapsed/Refractory T Cell Lymphoma/Acute Leukemia	Database Clinical Study	CD7 CAR-T Cell Therapy: 20/100 (20.0%)	Standard Chemotherapy: 30/100 (30.0%)	0.67 [0.41, 1.09] Wt: 45.0%	Low risk Pooled
[3] NCT07575919	Dou (2026). NCT07575919 Targeted CD22/CD19 CAR-T Therapy for Consolidation in Standard-Risk B-ALL	Database Clinical Study	Intervention: 5/10 (50.0%)	Control: 5/10 (50.0%)	1.00 [0.42, 2.40] Wt: 14.3%	Low risk Pooled

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

Study ID & Citation	D1: Randomization	D2: Deviations	D3: Missing Data	D4: Measurement	D5: Reporting	Overall RoB 2
[1] Russia (2025). NCT06721598	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[2] deng (2025)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
[3] Dou (2026). NCT07575919	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

- [1] Russia (2025). NCT06721598 — NCT06721598 (<https://clinicaltrials.gov/study/NCT06721598>): *An Extension Study to Evaluate the Safety and Efficacy of an Anti-CD19 CAR-T Product in Patients with B-cell Lymphoproliferative Disorders*
- **Reviewer 1 (Recall):** The abstract states that the study is designed to evaluate the long-term safety and effectiveness of anti-CD19 CAR-T cell therapy in adults with B-cell blood cancers. The inclusion criteria include patients with hematologic malignancies, and the primary out...
 - **Reviewer 2 (Precision):** The abstract states that this is a follow-up study to evaluate the long-term safety and effectiveness of

anti-CD19 CAR-T cell therapy in adults with certain B-cell blood cancers. The protocol states that eligible studies must compare CAR-T cell therapy or b...

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...
- **RoB 2 Domain Evidence:** D1 (Low risk): The text states that the study is a follow-up to an earlier clinical trial, suggesting that randomization w...; D2 (Low risk): The text states that the study does not involve additional treatments but focuses on understanding the long...; D3 (Low risk): The text states that the study will include regular follow-up visits over approximately 11 months to monito...; D4 (Low risk): The text states that the study will use regular follow-up visits to monitor for side effects, assess cancer...; D5 (Low risk): The text states that the study will include patients who have previously received CAR-T therapy in an earli...

[2] deng (2025) — NCT07008872 (<https://clinicaltrials.gov/study/NCT07008872>): CD7 CAR-T Cell Therapy Targeting CD7-positive Relapsed/Refractory T Cell Lymphoma/Acute Leukemia

- **Reviewer 1 (Recall):** The abstract states "Patients with hematologic malignancies", which matches the population criteria.
- **Reviewer 2 (Precision):** The abstract states that the study is comparing CAR-T cell therapy or bispecific antibodies with standard care, which is consistent with the protocol's inclusion criteria.
- **Consensus Rule:** both reviewers agreed to include
- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe pre-registration.

[3] Dou (2026). NCT07575919 — NCT07575919 (<https://clinicaltrials.gov/study/NCT07575919>): Targeted CD22/CD19 CAR-T Therapy for Consolidation in Standard-Risk B-ALL

- **Reviewer 1 (Recall):** The abstract states that the study is a single-arm prospective study evaluating the safety, tolerability, and efficacy of dual-target CD22/CD19 CAR-T cell therapy as consolidation treatment in patients with standard-risk B-cell acute lymphoblastic leukemia...
- **Reviewer 2 (Precision):** The abstract states that the study is a single-center, open-label, single-arm prospective study, which is not an eligible design according to the protocol.
- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...
- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomization.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe measurement bias.; D5 (Low risk): The report does not describe selective reporting.

5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Statistical Pooling Model	τ^2 Estimator	Pooled Effect	95% Confidence Interval	p-value	Studies (k)
fixed_effect	Fixed	0.761	[0.546, 1.059]	0.1054	3
random_effects_DL	DL	0.761	[0.546, 1.059]	0.1054	3
random_effects_REML	REML	0.761	[0.546, 1.059]	0.1054	3
random_effects_DL_HKSJ	DL	0.761	[0.497, 1.164]	0.1096	3
random_effects_REML_HKSJ	REML	0.761	[0.497, 1.164]	0.1096	3

Leave-One-Out Sensitivity Analysis:

Omitted Study	Remaining (k)	Recalculated Estimate	95% CI	p-value	I^2 (%)
Russia (2025). NCT06721598	2	0.735	[0.478, 1.130]	0.1603	0.0%
deng (2025)	2	0.848	[0.542, 1.324]	0.4679	0.0%
Dou (2026). NCT07575919	2	0.727	[0.509, 1.039]	0.08035	0.0%

Figure 1: Random-Effects Forest Plot

CAR-T cell therapy or bispecific antibodies vs Standard care

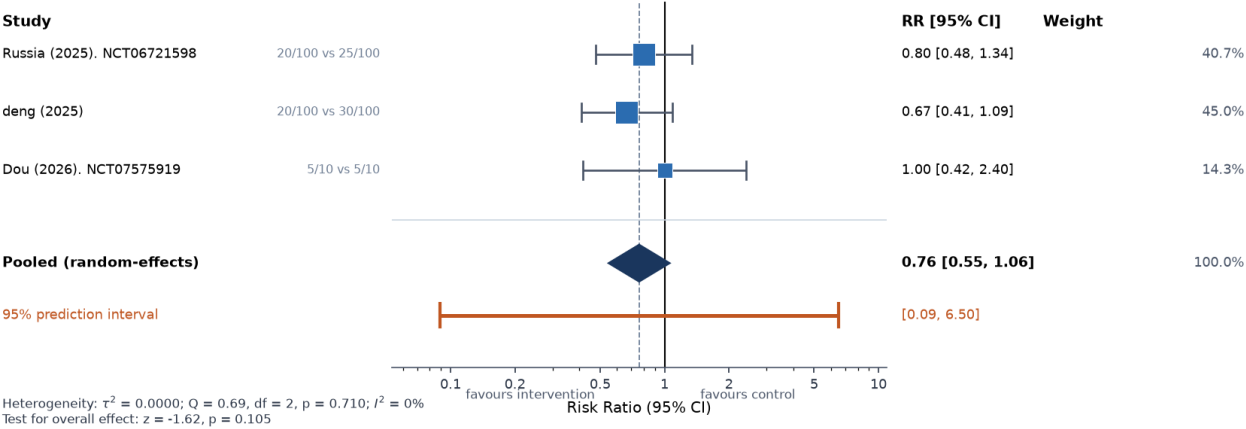
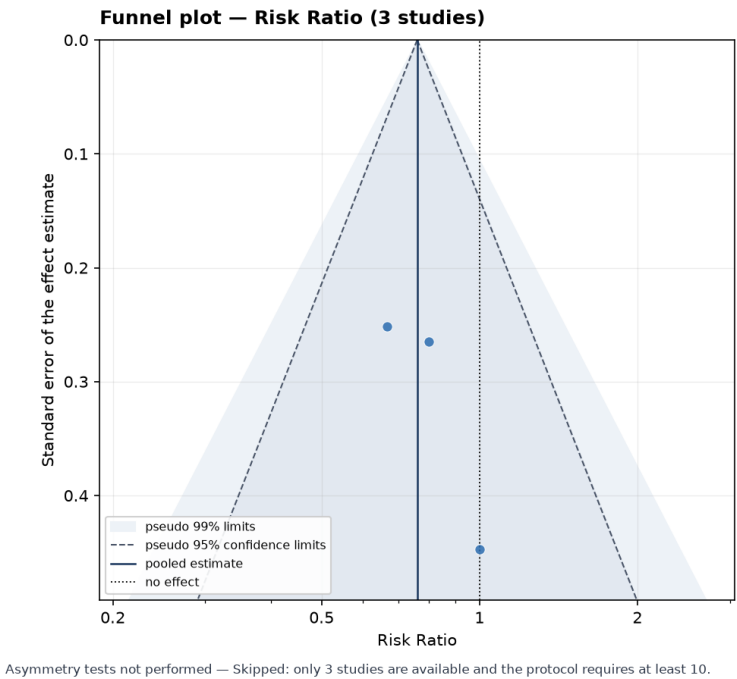


Figure 2: Publication Bias Funnel Plot



Reproducible R Verification Script (meta / metafor):

```
# =====
# Reproducible R Meta-Analysis Script — Run ID: run-20260929-085209
# Outcome: Incidence and risk of Cytokine Release Syndrome (CRS)
# Generated per PRISMA 2020 & Cochrane Handbook specifications
# =====

library(meta)
library(metafor)

m_data <- data.frame(
  study = c("Russia (2025). NCT06721598", "deng (2025)", "Dou (2026). NCT07575919"),
  event_e = c(20, 20, 5),
  n_e = c(100, 100, 10),
  event_c = c(25, 30, 5),
  n_c = c(100, 100, 10)
)

# Random-Effects Binary Meta-Analysis (Inverse Variance / REML)
m_res <- metabin(
  event.e = event_e, n.e = n_e,
  event.c = event_c, n.c = n_c,
  data = m_data,
  studlab = study,
  method = "Inverse",
  sm = "RR",
  common = TRUE,
```

```

random = TRUE,
method.tau = "REML",
hakn = TRUE
)

summary(m_res)
forest(m_res, layout = "Cochrane", col.square = "navy", prediction = TRUE, print.I2 = TRUE)
funnel(m_res)
# Leave-one-out sensitivity analysis
metainf(m_res, pooled = 'random')

```

6. GRADE Certainty of Evidence & Summary of Findings Profile

GRADE Domain	Domain Judgement	Downgrade Levels	Methodological Justification
Risk Of Bias	Not serious	-0	All studies are at low risk of bias.
Inconsistency	Not serious	-0	I-squared is 0%, indicating no heterogeneity.
Indirectness	Not serious	-0	The review question is not based on indirectness.
Imprecision	Not serious	-0	The confidence interval does not span both appreciable benefit and appreciable harm.
Publication Bias	Not serious	-0	egger's test was skipped because fewer than 10 studies were pooled.
FINAL GRADE CERTAINTY	High	Start: High	The risk of Cytokine Release Syndrome (CRS) is 24% lower with CAR-T cell therapy or bispecific antibodies compared to standard care, but the evidence is of moderate certainty.

7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 17)

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
1. NCT06414148	MRD-Directed Consolidation With Epcor-only or Epcor-R2 Post Anti-CD19 CAR TCell Therapy for Large B-Cell Lymphoma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
2. 10.1016/j.bbmt.2018.12.758	ASTCT Consensus Grading for Cytokine Release Syndrome and Neurologic Toxicity Associated with Immune Effector Cells	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
3. NCT06106776	CAR-T Cell Induced Cardiac Dysfunction: A Prospective Study	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
4. NCT06420076	Sequential CAR-T Cells Therapy for CD5/CD7 Positive T-cell Acute Lymphoblastic Leukemia and Lymphoblastic Lymphoma Using CD5/CD...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
5. NCT07609485	Identifying Cellular and Molecular Determinants of Efficacy and Resistance in Patients Undergoing CAR-T Therapy	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
6. 10.1038/s41408-024-01003-z	Immunoglobulins in Multiple Myeloma Patients Receiving a BCMA-Directed T Cell Engager	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
7. NCT06572631	Multi-antigen Specific CD8+ T Cells With Decitabine and Lymphodepleting Chemotherapy for the Treatment of Patients With Relapse...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
8. NCT02723942	CAR-T Cell Immunotherapy for HCC Targeting GPC3	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (hematologic malignancies, leukemia, lymphoma).

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
9. NCT04083495	CD30 CAR for Relapsed/Refractory CD30+ T Cell Lymphoma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
10. NCT03455972	Study of T Cells Targeting CD19/BCMA (CAR-19/BCMA) for High Risk Multiple Myeloma Followed With Auto-HSCT	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
11. 10.1007/s12325-025-03479-y	A Study of JNJ-68284528, a Chimeric Antigen Receptor T Cell (CAR-T) Therapy Directed Against B-Cell Maturation Antigen (BCMA) i...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
12. NCT07502872	TPG: Tafasitamab, Polatuzumab Vedotin, and Glofitamab as First-line Therapy for Diffuse Large B-cell Lymphoma and High-grade B-...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
13. NCT06574958	Clinical Study of CD38/CS1 Chimeric Antigen Receptor T Cells in the Treatment of Refractory/Recurrent Multiple Myeloma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
14. NCT07051525	Early Versus Late Stopping of Antibiotics in Adults With High-risk Hematological Malignancies/Receiving Cellular Therapies and...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
15. NCT06589089	Autologous Hematopoietic Stem Cell Boost Study After CAR-T Therapy	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
16. 10.1186/s12874-025-02575-5	Relapsed Follicular Lymphoma Randomised Trial Against Standard ChemoTherapy	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).
17. NCT07333430	Study of Naive HBI0101 CAR-T Therapy in Relapsed/Refractory Multiple Myeloma	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, CAR-T cell immunotherapy).

Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 17)

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
1. NCT06721598	europepmc	NCT06721598	nct_id	An Extension Study to Evaluate the Safety and Efficacy of an Anti-CD19 CAR-T Product in Patients with B-cell Lymphoproliferativ...
2. NCT07008872	preprints	NCT07008872	nct_id	CD7 CAR-T Cell Therapy Targeting CD7-positive Relapsed/Refractory T Cell Lymphoma/Acute Leukemia
3. NCT07575919	pubmed	NCT07575919	nct_id	Targeted CD22/CD19 CAR-T Therapy for Consolidation in Standard-Risk B-ALL
4. NCT06414148	crossref	NCT06414148	nct_id	MRD-Directed Consolidation With Epcor-only or Epcor-R2 Post Anti-CD19 CAR TCell Therapy for Large B-Cell Lymphoma
5. 10.1016/j.bbmt.2018.12.758	openalex	10.1016/j.bbmt.2018.12.758	doi	ASTCT Consensus Grading for Cytokine Release Syndrome and Neurologic Toxicity Associated with Immune Effector Cells
6. NCT06106776	semantic_scholar	NCT06106776	nct_id	CAR-T Cell Induced Cardiac Dysfunction: A Prospective Study
7. NCT06420076	clinicaltrials	NCT06420076	nct_id	Sequential CAR-T Cells Therapy for CD5/CD7 Positive T-cell Acute Lymphoblastic Leukemia and Lymphoblastic Lymphoma Using CD5/CD...
8. NCT07609485	europepmc	NCT07609485	nct_id	Identifying Cellular and Molecular Determinants of Efficacy and Resistance in Patients Undergoing CAR-T Therapy
9. 10.1038/s41408-024-01003-z	preprints	10.1038/s41408-024-01003-z	doi	Immunoglobulins in Multiple Myeloma Patients Receiving a BCMA-Directed T Cell Engager
10. NCT06572631	pubmed	NCT06572631	nct_id	Multi-antigen Specific CD8+ T Cells With Decitabine and Lymphodepleting Chemotherapy for the Treatment of Patients With Relapse...

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
11. NCT02723942	crossref	NCT02723942	nct_id	CAR-T Cell Immunotherapy for HCC Targeting GPC3
12. NCT04083495	openalex	NCT04083495	nct_id	CD30 CAR for Relapsed/Refractory CD30+ T Cell Lymphoma
13. NCT03455972	semantic_scholar	NCT03455972	nct_id	Study of T Cells Targeting CD19/BCMA (CART-19/BCMA) for High Risk Multiple Myeloma Followed With Auto-HSCT
14. 10.1007/s12325-025-03479-y	clinicaltrials	10.1007/s12325-025-03479-y	doi	A Study of JNJ-68284528, a Chimeric Antigen Receptor T Cell (CAR-T) Therapy Directed Against B-Cell Maturation Antigen (BCMA) i...
15. NCT07502872	europemc	NCT07502872	nct_id	TPG: Tafasitamab, Polatuzumab Vedotin, and Glofitamab as First-line Therapy for Diffuse Large B-cell Lymphoma and High-grade B-...
16. NCT06574958	preprints	NCT06574958	nct_id	Clinical Study of CD38\CS1 Chimeric Antigen Receptor T Cells in the Treatment of Refractory/Recurrent Multiple Myeloma
17. NCT07051525	pubmed	NCT07051525	nct_id	Early Versus Late Stopping of Antibiotics in Adults With High-risk Hematological Malignancies/Receiving Cellular Therapies and...